HEALTH-AI · ML Visualization Tool for Healthcare — Architecture Diagram

React 18 + Vite → FastAPI → scikit-learn

USER / BROWSER LAYER

Web Browser → 1 Clinical Context → 2 Data Exploration → 3 Data Preparation → 4 Model & Params → 5 Results → 6 Explainability → 7 Ethics & Bias

Domain Selector: Cardiology · Nephrology · Oncology · Neurology · Diabetes · Pulmonology · Sepsis/ICU · Fetal Health · Dermatology · Stroke

FRONTEND — React 18 + Vite (Deployed URL / localhost)

- Stepper Component — 7 interactive steps
- Domain Pill Bar — 10 healthcare domains
- CSV Upload Zone — Drag-drop + preview
- Parameter Sliders — K, C, max_depth
- Visualization Charts — Confusion matrix, KPI, ROC curve, bars
- Model Visualizations — KNN scatter, DT tree, SVM boundary
- Ethics & Bias Panel — EU AI Act checklist, Subgroup fairness
- Certificate Download (PDF)

Session Storage (Browser) — User selections, params, data, results

React State (useState / useReducer) — Stepper state, domain context, form data

HTTP Client (fetch / axios) — REST API calls to FastAPI

JSON

REST API ENDPOINTS (JSON over HTTP)

- POST /api/upload — CSV upload
- POST /api/preprocess — Clean & split
- POST /api/train — Train model
- POST /api/predict — Run prediction
- GET /api/results — Metrics
- GET /api/explain — Feature importance
- POST /api/certificate — Generate PDF

BACKEND — FastAPI (Python) · API docs via /docs

- API Router — FastAPI routes, Pydantic validation
- Data Service — CSV parsing, Missing values, Normalization, Train/Test split
- Training Service — Model selection, Hyperparameter config, Fit on train data
- Evaluation Service — Accuracy, Sensitivity, Specificity, AUC, Confusion matrix
- Explainability Service — Feature importance, Decision path trace, Patient-level reasoning
- Fairness / Bias Service — Subgroup analysis, EU AI Act checks, Disparity metrics
- Certificate Generator — PDF output

model.fit() / predict()

ML ENGINE — scikit-learn (Python) · 6 Classification Models

- KNN — K-Nearest Neighbors, Params: K, distance
- SVM — Support Vector Machine, Params: C, kernel
- Decision Tree — Tree-based splits, Params: max_depth
- Random Forest — Ensemble of trees, Params: n_estimators
- Logistic Regression — Linear classifier, Params: C, penalty
- Naive Bayes — Probabilistic classifier, Params: var_smoothing

scikit-learn Utilities: train_test_split · StandardScaler · confusion_matrix · accuracy_score · classification_report · roc_curve

DATA LAYER — CSV Datasets (No persistent DB — session-only processing)

- Default Datasets — Pre-loaded per domain (CSV)
- User-Uploaded CSV — Session-only, never stored
- Model Results Cache — In-memory (session)
- Generated Certificates — PDF download